

THINGSPEAK

This MATLAB script fetches 3-day weather forecasts from OpenWeatherMap, checks for expected rain, and sends the result to ThingSpeak for use in the Smart Agriculture irrigation system. It integrates with the ESP8266-based system to determine whether irrigation should be skipped due to predicted rain.

1. Overview

The script:

- Retrieves weather forecast data for a specific location (latitude: 36.8, longitude: 10.17) using the OpenWeatherMap API.
- Analyzes the next 72 hours (24 three-hour blocks) for precipitation.
- Sets a flag (rainExpected) to 1 if rain (≥ 0.1 mm) is predicted, otherwise 0.
- Sends the result to ThingSpeak (field 3) for the ESP8266 to read.

2. Code Structure

The script is written in MATLAB and uses web requests, JSON parsing, and ThingSpeak integration. It includes error handling to ensure reliability.

2.1 Configuration

- **API Keys and IDs:**
 - apiKey: OpenWeatherMap API key (845a9c746afdf739eaac68ec6e1bb979).
 - writeKey: ThingSpeak write API key (2OJMR1D255GTCJ24).
 - channelID: ThingSpeak channel ID (2906874).
- **Location:**
 - lat (36.8): Latitude for Tunis, Tunisia.
 - lon (10.17): Longitude for Tunis, Tunisia.

2.2 URL and Options

- **URL:**
 - Constructs an API request to OpenWeatherMap's forecast endpoint.
 - Format:
`https://api.openweathermap.org/data/2.5/forecast?lat={lat}&lon={lon}&appid={api Key}&units=metric.`
 - Requests data in metric units (e.g., precipitation in mm).
- **Options:**

- weboptions: Sets content type to text and a 30-second timeout for the API request.

2.3 Main Logic

- **Fetch and Parse Data:**
 - Uses webread to retrieve JSON data from OpenWeatherMap.
 - Parses JSON into a MATLAB structure using jsondecode.
 - Checks if the list field (containing forecast data) exists and is valid; otherwise, throws an error.
- **Rain Check:**
 - Iterates through up to 24 three-hour forecast blocks (covering 72 hours).
 - For each block:
 - Extracts the timestamp (dt_txt) if available.
 - Checks for precipitation (rain.3h field, representing mm of rain in 3 hours).
 - If precipitation ≥ 0.1 mm, sets rainExpected = 1 and stops checking.
 - Logs timestamp and precipitation (or lack thereof) to the console.
- **ThingSpeak Update:**
 - Sends rainExpected (0 or 1) to ThingSpeak's field 3 using thingSpeakWrite.
 - Logs the result to the console.

2.4 Error Handling

- Wrapped in a try-catch block to handle errors (e.g., API failures, invalid JSON).
- Displays error messages if the request or parsing fails.

3. Key Features

- **Weather Integration:** Fetches and analyzes 72-hour forecasts to predict rain.
- **Simple Logic:** Sets a binary flag (rainExpected) for easy use by the ESP8266.
- **Reliable Communication:** Updates ThingSpeak for integration with the irrigation system.
- **Error Handling:** Ensures robustness by catching and logging errors.

4. Limitations and Improvements

- **Hardcoded Location:** Latitude and longitude are fixed; could be made configurable.
- **Rain Threshold:** Fixed at 0.1 mm; could be adjustable based on crop needs.
- **Future Enhancements:**
 - Add support for multiple locations.
 - Include additional weather parameters (e.g., temperature).
 - Log errors to a file for debugging.

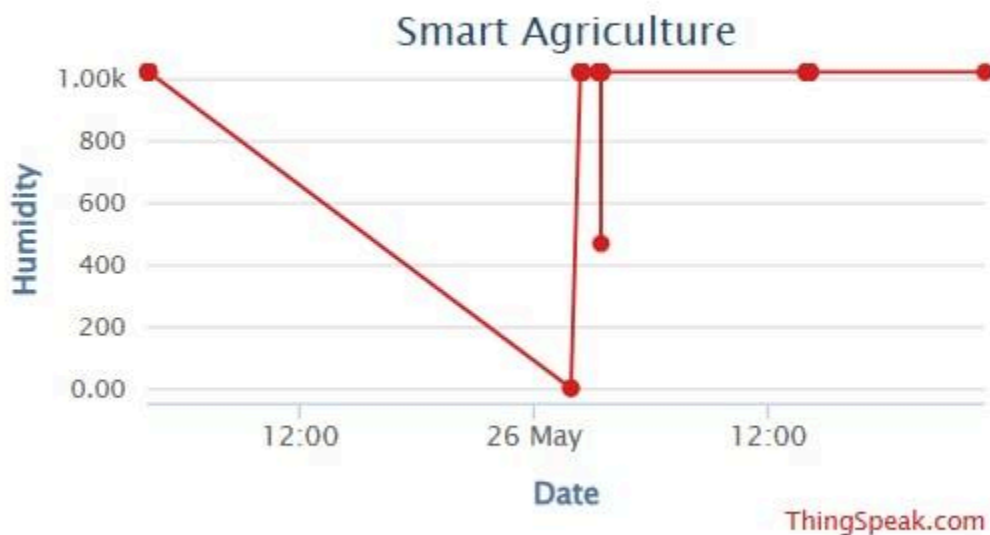
5. Conclusion

This MATLAB script provides a simple and reliable way to integrate weather forecasts into the Smart Agriculture system. By checking for rain and updating ThingSpeak, it enables the ESP8266 to make informed irrigation decisions, reducing water waste.

Output

```
2025-05-28 12:00:00 - Aucune donnée de pluie.  
2025-05-28 15:00:00 - Aucune donnée de pluie.  
2025-05-28 18:00:00 - Aucune donnée de pluie.  
2025-05-28 21:00:00 - Aucune donnée de pluie.  
2025-05-29 00:00:00 - Aucune donnée de pluie.  
2025-05-29 03:00:00 - Aucune donnée de pluie.  
2025-05-29 06:00:00 - Aucune donnée de pluie.  
2025-05-29 09:00:00 - Aucune donnée de pluie.  
2025-05-29 12:00:00 - Aucune donnée de pluie.  
2025-05-29 15:00:00 - Aucune donnée de pluie.  
2025-05-29 18:00:00 - Aucune donnée de pluie.  
2025-05-29 21:00:00 - Aucune donnée de pluie.  
???? Résultat envoyé à ThingSpeak (Field 3) : 0
```

Field 1 Chart



ThingSpeak.com

Field 3 Chart



Smart Agriculture

RainExpected

0.00



Date

ThingSpeak.com